

Question:1

Write a C program to print all natural numbers in reverse from n to 1 using for loop

```
#include<stdio.h>

int main()
{
    int n, i;
    printf("Enter the number you want to print from:\t");
    scanf("%d",&n);
    for(i=n; i>=1;i--)
        printf("%d\n",i);
    return 0;
}
```

```
Enter the number you want to print from:      20
20
19
18
17
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1

-----
Process exited after 2.223 seconds with return value 0
Press any key to continue . . .
```

Question:2

Write a C program that user performs multiplication and division without using the * and / operators:

```
#include <stdio.h>

int main() {
    int a, b, i, mul = 0, div = 0, c;
    printf("Enter first number:\t");
    scanf("%d", &a);
    printf("Enter second number:\t");
    scanf("%d", &b);
    for (i = 0; i < b; i++) {
        mul = mul + a;
    }
    c = a;
    while (c >= b) {
        c = c - b;
        div++;
    }
    printf("\nMultiplication = %d", mul);
    printf("\nDivision = %d", div);
    return 0;
}
```

```
Enter first number:    20
Enter second number:   5

Multiplication = 100
Division = 4
-----
Process exited after 3.807 seconds with return value 0
Press any key to continue . . .
```

Question:3

Write a C program to input a number and find sum of first and last digit of the number using for loop.

```
#include <stdio.h>

int main() {
    int num, first, last, sum;
    printf("Enter a number:\t");
    scanf("%d", &num);
    last = num % 10;
    for (; num >= 10; num = num / 10) {
    }
    first = num;
    sum = first + last;
    printf("Sum of first and last digit = %d", sum);
    return 0;
}
```

```
Enter first number:    20
Enter second number:   5

Multiplication = 100
Division = 4
-----
Process exited after 3.807 seconds with return value 0
Press any key to continue . . .
```

Question:4

Write a C program that takes an integer input from the user and checks whether It is a prime number or not.

A prime number is a natural number greater than 1. That has no positive divisors other than 1 and itself.

Example:

- Input: 17 → Output: "17 is a prime number"
- Input: 15 → Output: "15 is not a prime number"

```
#include<stdio.h>

int main()
{
    int a,num;
    printf("Enter the number you want to check:\t");
    scanf ("%d",&num);
    for (a=2;a<num;a++)
        if (num%a==0)
            break;
    if (a==num)
        printf("\nThe number is a prime number");
    else
        printf("\nThe number is not a prime number");
    return 0;
}
```

```
Enter the number you want to check:    17
The number is a prime number
-----
Process exited after 2.051 seconds with return value 0
Press any key to continue . . .
```

```
Enter the number you want to check:    15
The number is not a prime number
-----
Process exited after 1.08 seconds with return value 0
Press any key to continue . . .
```

Question:5

Write a C program that asks the user for 10 integer values and counts how many are even and how many are odd.

```
#include<stdio.h>

int main()
{
    int number, i, even=0, odd=0;
    for (i=1;i<=10;i++)
    {
        printf("Enter %d number:", i);
        scanf ("%d",&number);
        if (number%2==0)
            even++;
        else
            odd++;
    }
    printf("You entered %d even numbers:\t",even);
    printf("You entered %d odd numbers:\t",odd);
    return 0;
}
```

```
Enter 1 number:0
Enter 2 number:1
Enter 3 number:2
Enter 4 number:3
Enter 5 number:4
Enter 6 number:5
Enter 7 number:6
Enter 8 number:7
Enter 9 number:8
Enter 10 number:9
You entered 5 even numbers:      You entered 5 odd numbers:
-----
Process exited after 7.14 seconds with return value 0
Press any key to continue . . .
```

Question: 6

Write a C program that takes an integer input from the user and calculates the sum of its individual digits.

Example:

- Input: 1234 → Output: "Sum of digits: 1+2+3+4 = 10"
- Input: 987 → Output: "Sum of digits: 9+8+7 = 24"

```
#include <stdio.h>

int main() {
    int num, a, digit, sum = 0, rev = 0;
    printf("Enter an integer: ");
    scanf("%d", &num);
    a = num;
    while (a != 0) {
        rev = rev * 10 + (a % 10);
        a = a / 10;
    }
    printf("Sum of digits: ");
    while (rev != 0) {
        digit = rev % 10;
        sum += digit;
        rev = rev / 10;
        if (rev != 0)
            printf("%d+", digit);
        else
            printf("%d", digit);
    }
    printf(" = %d", sum);
    return 0;
}
```

```
Enter an integer: 1234
Sum of digits: 1+2+3+4 = 10
-----
Process exited after 1.862 seconds with return value 0
Press any key to continue . . .
```

```
Enter an integer: 987
Sum of digits: 9+8+7 = 24
-----
Process exited after 2.336 seconds with return value 0
Press any key to continue . . .
```

Question:7

Write a C program that takes an integer input from the user and displays the number in reverse order.

Example:

- Input: 12345 → Output: 54321
- Input: 100 → Output: 001.

```
#include <stdio.h>
int main()
{
    int Num, rev_Num = 0, remainder;
    printf("Enter the number to reverse: ");
    scanf("%d", &Num);
    while (Num != 0) {
        remainder = Num % 10;
        rev_Num = rev_Num * 10 + remainder;
        Num = Num/10;
    }
    printf("The reversed number is: %d", rev_Num);
    return 0;
}
```

```
Enter the number to reverse: 12345
The reversed number is: 54321
-----
Process exited after 10.97 seconds with return value 0
Press any key to continue . . .
```

Question:8

Write a C program to check whether a given number is a perfect number. A perfect number is a positive integer that is equal to the sum of its proper positive divisors, excluding the number itself

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#include <stdio.h>

```
int main() {
    int num, i, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    for (i = 1; i < num; i++) {
        if (num % i == 0) {
            sum = sum + i;
        }
    }
    if (sum == num)
        printf("%d is a Perfect Number.", num);
    else
        printf("%d is NOT a Perfect Number.", num);
    return 0;
}
```

```
Enter a number: 125
125 is NOT a Perfect Number.
-----
Process exited after 3.346 seconds with return value 0
Press any key to continue . . .
```

```
Enter a number: 6
6 is a Perfect Number.
-----
Process exited after 0.8203 seconds with return value 0
Press any key to continue . . .
```

Question:9

Sarah wants to track her monthly expenses more efficiently. She needs a program that can handle multiple expense categories and provide detailed financial analysis

Program Requirements:

1. Takes Sarah's monthly income as input
2. Collects expenses for 4 categories: **Food, Transport, Entertainment, Utilities**
3. Calculates and displays:
 - Total monthly expenses
 - Remaining balance after expenses

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- Savings amount and what percentage this represents of her income
- The percentage that each expense category contributes to total expenses

4. Provides personalized financial advice:

- If her savings are more than 20% of income: "Excellent savings!"
- If any single expense category makes up more than 40% of total expenses: "Warning: High spending in [category]"
- If she spends more than her income: "You are overspending!"

```
#include<stdio.h>

int main()
{
    float income, food, transport, entertainment, utilities, expences, save, remaining, saving, fexpance, texpance, eexpance, uexpance;
    printf("Enter your monthly income:\t");
    scanf("%f",&income);
    printf("Enter Food Expense:\t");
    scanf("%f",&food);
    printf("Enter Transport Expense:\t");
    scanf("%f",&transport);
    printf("Enter Entertainment Expense:\t");
    scanf("%f",&entertainment);
    printf("Enter Utilities Expense:\t");
    scanf("%f",&utilities);
    expences=food+transport+entertainment+utilities;
    remaining=income-expences;
    save=remaining;
    saving=(remaining/income)*100;
    fexpance=(food/expences)*100;
    texpance=(transport/expences)*100;
    eexpance=(entertainment/expences)*100;
    uexpance=(utilities/expences)*100;
    printf("Total monthly expenses:\t%.2f",expences);
    printf("\nRemaining balance after expenses:\t%.2f",remaining);
    printf("\nSavings amount :\t%.2f",save);
    printf("\nSavings amount percentage :\t%.2f%% of income",saving);
    printf("\nFood Expance percentage :\t%.2f%% of income",fexpance);
    printf("\nTransport Expance percentage :\t%.2f%% of income",texpance);
    printf("\nEntertainment Expance percentage :\t%.2f%% of income",eexpance);
    printf("\nUtility Expance percentage :\t%.2f%% of income",uexpance);
    if(saving>20)
        printf("\nExcellent Saving!");
    if(fexpance>40)
        printf("\nWarning: High spending in Food Expance");
    else if(texpance>40)
        printf("Warning: High spending in Transport Expance");
    else if(eexpance>40)
        printf("Warning: High spending in Transport Expance");
    else if(uexpance>40)
        printf("Warning: High spending in Utility Expance");
    if(expences>income)
        printf("You are overspending!");
    return 0;
}
```

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```
Enter your monthly income: 100000
Enter Food Expanse: 35000
Enter Transport Expanse: 15000
Enter Entertainment Expanse: 10000
Enter Utilities Expanse: 8000
Total monthly expenses: 68000.00
Remaining balance after expenses: 32000.00
Savings amount : 32000.00
Savings amount percentage : 32.00% of income
Food Expanse percentage : 51.47% of income
Transport Expanse percentage : 22.06% of income
Entertainment Expanse percentage : 14.71% of income
Utility Expanse percentage : 11.76% of income
Excellent Saving!
Warning: High spending in Food Expanse
-----
Process exited after 23.58 seconds with return value 0
Press any key to continue . . .
```